**TASK: 1**

**Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.**

**Aim:** To create and configure two VLANs in Cisco Packet Tracer.

**Commands:**

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit
Switch(config)# end
Switch# show vlan brief
```

**Expected Output:**

```
VLAN Name                 Status    Ports
---- -------------------- --------- ----------------
1    default              active    Fa0/6, Fa0/7...
10   Sales                active
20   Support              active
```

**Result:** Two VLANs were created successfully: VLAN 10 named Sales and VLAN 20 named Support.

NETWORK

**TASK: 2**

**Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.<br><br>Hint: Use the interface range command to select multiple ports efficiently.**

**Aim:** To assign three switch ports to Sales and two ports to Support.

**Commands:**

**Assign ports Fa0/1–Fa0/3 to VLAN 10 (Sales):**

```
Switch> enable
Switch# configure terminal
Switch(config)# interface range fastEthernet 0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

**Assign ports Fa0/4–Fa0/5 to VLAN 20 (Support):**

```
Switch(config)# interface range fastEthernet 0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
Switch(config)# end
Switch# show vlan brief
```

**Expected Output:**

```
VLAN Name                 Status    Ports
---- -------------------- --------- ----------------
1    default              active    Fa0/6, Fa0/7...
10   Sales                active    Fa0/1, Fa0/2,
Fa0/3
20   Support              active    Fa0/4, Fa0/5
```

**Result:** Three ports were assigned to Sales (VLAN 10), and two ports were assigned to Support (VLAN 20). The `show vlan brief` command verifies the VLAN membership.

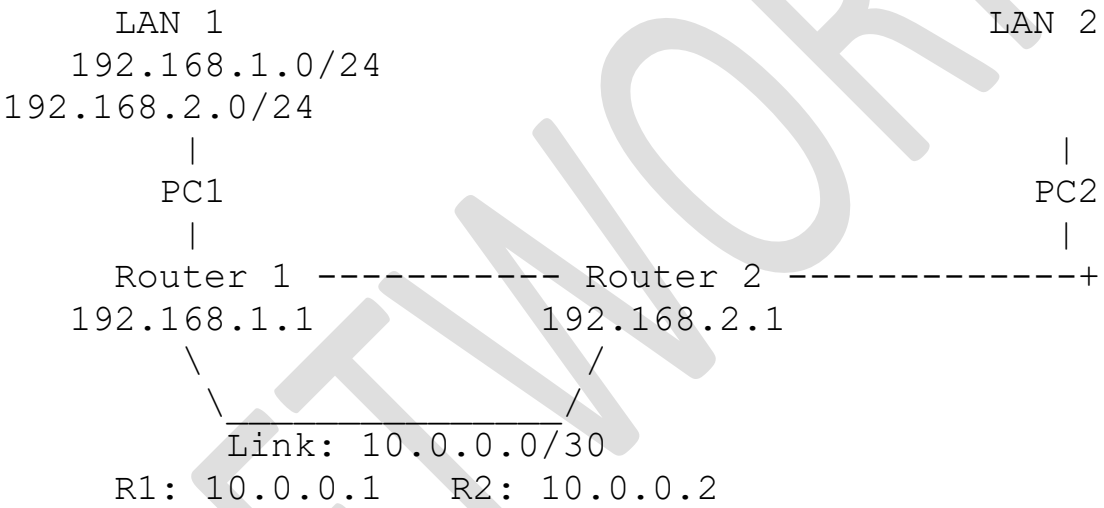
NETWORK

**TASK: 3**

## Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router.

**Aim:** To connect two routers and configure static routing so Router 1 can reach the LAN behind Router 2.

**Network Diagram**




**Router 1 Configuration**

Assume Router 1 uses GigabitEthernet0/0 for its LAN and GigabitEthernet0/1 for the link to Router 2.

```
Router1> enable
Router1# configure terminal

Router1(config)# interface gigabitEthernet 0/0
Router1(config-if)# ip address 192.168.1.1
255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit

Router1(config)# interface gigabitEthernet 0/1
```

NETWORK

```
Router1(config-if)# ip address 10.0.0.1
255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit

Router1(config)# ip route 192.168.2.0 255.255.255.0
10.0.0.2
Router1(config)# end
Router1# show ip route
```

**Router 2 Configuration**

```
Router2> enable
Router2# configure terminal

Router2(config)# interface gigabitEthernet 0/0
Router2(config-if)# ip address 192.168.2.1
255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)# exit

Router2(config)# interface gigabitEthernet 0/1
Router2(config-if)# ip address 10.0.0.2
255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit

Router2(config)# ip route 192.168.1.0 255.255.255.0
10.0.0.1
Router2(config)# end
Router2# show ip route
```

**Verification:** Use `ping 10.0.0.2` from Router 1 to test the inter-router connection. Once the interfaces, routes, and PC default gateways are configured correctly, test connectivity between the PCs.

**Result:** Static routes allow Router 1 to reach the `192.168.2.0/24` network through Router 2. Router 2 also has a return route to `192.168.1.0/24`.

NETWORK

**TASK: 4**

## Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.<br><br>Hint: Think about router-on-a-stick or Layer 3 switch concepts.

**Scenario:** A computer connected to Sales (VLAN 10) tries to ping a computer connected to Support (VLAN 20).

**Explanation:**

When a device in VLAN 10 tries to ping a device in VLAN 20, the ping normally fails if only Layer 2 switching is configured. VLANs create separate broadcast domains, and a Layer 2 switch does not route traffic between them by default. Devices in different VLANs require a Layer 3 routing function to communicate.

**Solution: Configure Inter-VLAN Routing**

Inter-VLAN communication can be enabled using either of these methods:

1. **Router-on-a-stick:** Connect a router to the switch through a trunk link and configure router subinterfaces for VLANs 10 and 20. Each subinterface receives an IP address that acts as the default gateway for its VLAN.
2. **Layer 3 switch:** Configure switched virtual interfaces (SVIs) for VLANs 10 and 20 and enable IP routing on the switch.

For router-on-a-stick, the switch port connected to the router must be configured as a trunk, and the router must have correctly configured subinterfaces using 802.1Q encapsulation.

**Example addressing:**

NETWORK

| VLAN | Name | Network | Default Gateway |
|---|---|---|---|
| 10 | Sales | 192.168.10.0/24 | 192.168.10.1 |
| 20 | Support | 192.168.20.0/24 | 192.168.20.1 |

**Result:** Initially, ping fails because the VLANs are isolated at Layer 2. After configuring inter-VLAN routing, correct IP addresses, and default gateways, devices in Sales and Support can communicate, provided that no ACL or firewall rule blocks the traffic.

NETWORK